4. Dry heat:

Sterilizing of glassware including syringes is often done in a hot air oven at 160°C for one hour. Spores as well as organisms are killed. Rubber will not stand up to this heat. This method is effective for dressing towels and gowns.

Two methods are used:

1. **Incineration:** Used for disposal of dressings laboratory media and human tissues. Incinerator may cause unacceptable environmental pollution.
2. **Hot air oven:** It is used for objects which can not tolerate moisture. Mostly used for glass ware, oils and powders.

Disinfection: Destruction of organisms by chemical is used in the following circumstances.

1. **Environmental:** Disinfection of excreta, floors, furniture, linen and fabrics.
2. **Instruments/Equipments:** Sterilization of heat sensitive objects in contact with patient.
3. **Skin and Wounds:** Removal of pathogens.
4. **Food medication:** Preservation in prevention of spoilage.
5. **Water:** Removal of pathogens.

5.2.2 Chemical sterilization

It is the method used for eye instruments and other delicate instruments.

Points to remember:

- Articles must be clean and free from pus, blood or oil.
- It must be completely immersed in the disinfectant.
- The disinfectant should be of a certain strength and articles must be in contact with it for a certain length of time.
- After sterilization articles must be well rinsed in sterile water before use.

Types of chemicals:

A Alkylating agents:

They are capable of killing bacteria, spores and viruses. They are the acceptable chemical alternative to usual heat treatment. This include:

1. **Formaldehyde:** Used for instruments and machines sterilization.
2. **Gluteraldehyde:** Used for instrument sterilization.
3. **Ethylene oxide:** Used for rubber and plastic articles sterilization.
4. **Propiolactone:** Used for gaseous sterilization.

B Phenols:

It includes acids and semi synthetic compounds in soap solution. It is used for contaminated surfaces.

Advantages:

This is the method used to sterilize the articles that are destroyed by heat.

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